mucus (Figure 9.2). Droplets are relatively large, about 10µm or more in size, and they tend to settle rapidly in still air. When inhaled these droplets are trapped on the moist surfaces of the respiratory tract. Thus, the droplets containing pathogenic microorganisms may be a source of infectious disease.



Figure 9.2: Droplets released during sneezing

Droplet nuclei

Small droplets in a warm, dry atmosphere tend to evaporate rapidly and become droplet nuclei. These are 1-4µm in size. They can remain in air for longer period and transmit various infectious airborne diseases.

Infectious Dust

Large aerosol droplets settle out rapidly from air on to various surfaces and get dried. Nasal and throat discharges from patient can also contaminate surfaces and become dry. Microorganisms can survive for longer period in dust. This creates a significant hazard, especially in hospital areas.

9.2.5 Air Borne Diseases

Many microbial diseases are transmitted through the air (Table 9.2). The incidence of diseases caused by airborne transmission can be reduced by covering one's nose and mouth during coughing or sneezing and by the use of face masks.

Table 9.2: Important airborne human diseases and causative agents (pathogens)

Human Diseases	Pathogens
Bacterial diseases	
Pulmonary tuberculosis	*Mycobacterium tuberculosis*
Pneumonia	*Klebsiella pneumoniae*
Streptococcal respiratory infections	*Streptococcus pyogenes*
Fungal diseases	
Aspergillosis	*Aspergillus fumigatus*
Cryptococcosis	*Cryptococcus neoformans*
Viral diseases	
Influenza	*Influenza Virus*
Common cold	*Picorna Virus*
Protozoal diseases	
Pneumocystosis	*Pneumocystis carinii*

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